"INCH-POUND"
MIL-D-2457J
1 November 1988
SUPERSEDING
MIL-D-2457H
11 March 1977

MILITARY SPECIFICATION

DUSTER, INSECTICIDE, MASS DELOUSING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of dusters used for mass delousing of personnel and clothing.

1.2 Classification. The Mass Delouser shall be of the following types, as specified (see 6.2).

Type I - Air supply unit, capable of 4.2 C.F.M., and accessory box with all equipment, total weight not to exceed 141 pounds (see 3.4.1).

Type II - For units that have organic air supplies, accessory box with all equipment, total weight not to exceed 70 pounds (see 3.4.2).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Biomedical Research and Development Laboratory, ATTN: SGRD-UBE-D, Fort Detrick, Frederick, MD 21701-5010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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by suppliers in connection with specific procurement functions should be

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

TT-E- 529 - Enamel, Alkyd, Semigloss.
TT-P-1757 - Primer Coating, Zinc Chromate, Low Moisture Sensitivity.
PPP-T-60 - Tape, Packaging, Waterproof.

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MIL-P-116 - Preservation, Method of.
MIL-B-121 - Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-S-1837 - Spray Gun, Insecticide Dusting Powder and Container, Powder.
MIL-R-3308 - Respirator, Air Filtering, Dust, ABC-M4.
MIL-E-10062 - Engine, Preparation for Shipment and Storage of.
MIL-C-10578 - Corrosion Removing and Metal Conditioning Compound (Phosphoric Acid Base).
MIL-C-16775 - Chest, Field, Aluminum, Empty.
MIL-G-23827 - Grease, Aircraft, and Instrument, Gear, and Actuator Screw.
MIL-M-38784 - Manuals, Technical: General Style and Format Requirements.
MIL-T-46786 - Tank, Fuel, Engine General Requirements.
MIL-C-52950 - Crates, Wood, Open and Covered.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.
FED-STD-595 - Colors.
FED-STD-601 - Rubber: Sampling and Testing.

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MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

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MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of U.S. Military Property.
MIL-STD-1226 - Engine, Gasoline: Air-Cooled, 1 1/2 BHP 4-Cycle, Military Design, Model 1A08 Installation.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

U.S. ARMY BIOMEDICAL RESEARCH AND DEVELOPMENT LABORATORY

PL 60107FD - Assembly and Parts List of Delousing Gun.
60107FD - Duster, Insecticide, Mass Delousing: Assembly.
60108FD - Duster, Insecticide, Mass Delousing: Box, Parts, Shipping and Storage.
60109FD - Duster, Insecticide, Mass Delousing: Manifold Assembly and Miscellaneous Details.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Commercial Packaging.

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These

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documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the reference cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The Duster, Insecticide, Mass Delousing shall be in accordance with drawings PL 60107FD, 60107FD, 60108FD, and 60109FD and as specified herein. No deviation from the prescribed dimensions or tolerances is permissible without prior approval to ensure correct fit, assembly, and operation of the production item. Unless otherwise specified in the contract or purchase order, data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the supplier or obtained from a vendor to support fabrication and manufacture of the production item shall be made available for inspection by the contracting officer or his designated representative. In the event of any conflict between the requirements of this specification and the drawings, the latter shall govern (see 4.2).

3.2 First article. When specified in the contract or purchase order, two complete units shall be subjected to first article inspection (see 4.3 and 6.4). Examinations and tests shall be as specified herein and during the time frame specified; and unless otherwise specified, all examinations and tests shall be conducted by the Government.

3.3 Identical items. All delousers furnished under a specific contract shall be physically and mechanically identical; all parts, components, accessories, and subassemblies shall be interchangeable on all delousers (see 4.1). No deviation shall be acceptable without prior written approval from the contracting officer.

3.4 Materials and components. The materials and components shall be as specified herein and on the drawings. Materials and components not definitely specified shall be of the quality normally used by the manufacturer for insecticide dusters, provided the complete item complies with all provisions of this specification (see 4.1).

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3.4.1 Type I Delouser. The type I duster shall consist of the following components as specified herein:

Quantity per outfit	Nomenclature
1	Compressor Unit, Power Driven Gasoline Engine, Duster (see 4.5.2 and 4.5.4).
1	Box, Parts, Shipping and Storage, Duster (see 4.1).
6	Spray Gun, Including Container (see 4.1).
6	Hose, 1/4-inch (6.4 mm) ID, 25 feet (7.62 m) long, Complete with Couplers (see 4.4.1.1).
1	Manifold, Complete with Air Brake Connection (see 4.5.1 and 4.5.2).
12	Respirator, Air Filtering, M4 (see 6.2).
4	Latches (see 4.1).
1	Operator's, Organizational, Direct and General Support Maintenance Manual (see 4.1 and 6.3).

3.4.2 Type II Delouser. The type II duster shall consist of the following components as specified herein.

Quantity per outfit	Nomenclature
1	Box, Parts, Shipping and Storage, Duster (see 4.1).
6	Spray Gun, Including Containers (see 4.1).
6	Hose, 0.25 inch (6.4 mm) ID, 25 feet. (7.62 m) long, Complete with Couplers (see 4.4.1.1).
1	Manifold, Complete with Air Brake Connection (see 4.5.1 and 4.5.2).

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3.4.2 (cont'd)

Quantity per outfit	Nomenclature
12	Respirator, Air Filtering, M4 (see 6.2).
4	Latches (4.1).
1	Operator's, Organizational, Direct and General Support Maintenance Manual (see 4.1 and 6.3).

3.4.3 Bolts, studs, nuts, and washers. Bolts, studs, nuts, and washers used for mounting components, but not specified, shall be the sizes and types recommended by the contractor and shall conform to current commercial standards. Threads shall be class 2 fit and shall conform to FED-STD-H28. Steel bolts, studs, nuts, and washers shall be zinc plated.

3.4.4 Spray guns. Spray guns, complete with plastic containers, shall conform to MIL-S-1837.

3.4.5 Grease, bearing. Bearing grease shall conform to the requirements of MIL-G-23827.

3.4.6 Primer. Primer for all metal surfaces shall conform to TT-P-1757.

3.4.7 Enamel. Enamel shall be semi-gloss conforming to TT-E-529. The color shall match Forest Green Type II color No. 34083 of FED-STD-595.

3.4.8 Government furnished equipment. The M4 air filtering respirators shall be Government furnished and shall conform to the requirements of MIL-R-3308 (see 6.2).

3.5 Design. The duster shall consist of an air compressor driven by a gasoline engine and mounted on a common base enclosed within an aluminum tubular frame. The compressor and engine shall be connected by a timing belt drive. A manifold shall be equipped with six quick-disconnect air hose coupling assemblies, a pressure gauge, a drain cock, a safety relief valve, and an air brake hose connection (see 3.4.1). An aluminum parts box (No. 2 Medical Chest), detachable from the metal frame, shall be furnished for storing air hoses, spray guns, spare plastic containers, dust respirators, tools and accessories (3.6.2). The net weight of the type I Delouser with gasoline engine, parts box, hoses, and all accessories shall not exceed 141 pounds (64.0 kg). The net weight of the type II Delouser with parts box, hoses, and all accessories shall not exceed 70 pounds (31.75 kg) (see 4.1).

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3.6 Construction. The construction of the duster shall conform to the requirements specified herein and on the drawings. The finished duster shall be capable of operating without visible signs of leaks, breakdown, shock, vibration, or wear; when tested as specified in 4.5.1, 4.5.2, 4.5.3, and 4.5.4, as applicable. When subjected to endurance test specified in 4.5.4, there shall be no significant change in the performance characteristics recorded as specified therein.

3.6.1 Engine, gasoline. The gasoline engine shall be 4-cycle, capable of producing not less than 1.5 net continuous brake horsepower at 3,600 revolutions per minute (rpm), radio suppressed, conforming to MIL-STD-1226. All serviceable components of the engine such as air cleaner, magneto, spark plug, gasoline strainer, and carburetor shall be easily accessible for service in the engine compartment without removal of the engine, or any other components of the duster. The following specific accessories are considered essential to the satisfactory performance of the gasoline engine and shall be included with the standard specified accessories and auxiliaries (see 4.5.4).

3.6.1.1 Engine grounding. The installed gasoline engine shall be securely grounded to the frame base with a grounding strap or with flexible wire not less than size No. 12 (see 4.1).

3.6.1.2 Engine governor. The engine shall be governed at the required speed to operate the compressor at its rated capacity. There shall be a hand operated attachment for adjusting to the required speed (see 4.1).

3.6.1.3 Fuel tank. The fuel tank shall conform to the requirements of MIL-T-46786 and shall have a capacity of not less than 1 gallon (3.79 liters).

3.6.2 Compressor, air. The compressor shall be oil-less in design. The compressor shall be equipped with a safety valve. The discharged air from the compressor shall show no evidence of hydrocarbon when tested as specified in 4.5.4. The use of exhaust filters or other similar attachment devices shall not be permitted. The inlet to the air compressor shall be provided with a removable dry-type air filter. The compressor shall not leak when tested in accordance with 4.5.2.

3.6.2.1 Capacity. The air compressor, when driven by the gasoline engine specified in 3.6.1, shall deliver 4.2 cubic feet of oil-free air per minute (0.0020 m^3/s) with manifold pressure of 30 pounds per square inch (207 kPa) gauge (psig) when tested as specified in 4.5.5.

3.6.2.2 Timing belt. When a timing belt is used, it shall require the minimum distance between the shaft centerlines. Standard mating timing pulleys shall be used to obtain the proper speed reduction (see 4.1).

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3.6.3 Manifold. The manifold shall have six female quick disconnect couplers, a 0-100 psig pressure gauge, a pressure relief valve, a drain cock, and a securing strap. The manifold shall include a truck air brake connection and female quick disconnect coupler on opposite ends of enough air hose (see 3.5.4) to equal 30 inches. The manifold shall not leak when tested as specified in 4.5.1.

3.6.3.1 Gauge, pressure, oil-filled. The gauge shall have a 1/4-inch National Pipe Thread (N.P.T.) male back connection. The dial shall be 2 inch (150.8 mm) size constructed with white background and black markings to read from 0 to 100 psig. The gauge shall be oil-filled to withstand vibration.

3.6.3.2 Valve, safety. The safety valve shall have a 1/4-inch N.P.T. male connection and shall be preset to discharge at 30 psig (207 kPa) when tested as specified in 4.5.4. The safety valve shall have a pressure rating of 150 psig (1034 kPa).

3.6.4 Hoses with couplings. Each duster shall be provided with six spray hoses, each 25 feet (7.6 m) long. Each spray hose shall be fitted with hose couplings. One end of the hose shall have a type II, class A, style 3 half coupling, and the other end shall have a type I, class A, style 3 half coupling attached. The rubber hose shall be commercial grade, single braid, designed for use with compressed air. The hose shall be 1/4-inch (6.4 mm) I.D. with a black cover. Minimum burst pressure shall be 800 psig (5520 kPa), and the ultimate elongation for the tube shall be not less than 250 percent and the cover 350 percent when tested as specified in 4.4.1.1.

3.6.5 Spray guns and plastic containers. Spray guns and plastic containers shall conform to 3.4.4 and shall be provided in the quantities specified in 3.4.1 or 3.4.2.

3.6.6 Framework and mounting. The entire duster shall be mounted within an aluminum frame. Mounting of the engine compressor assembly shall be on an aluminum base with rubber bushings to absorb vibrations. Torsional vibration induced at rated output of the compressor when coupled to the engine shall be minimized by dynamic balancing of the compressor. The base shall be fabricated from aluminum.

3.6.7 Parts box. The parts box shall be a standard No. 2 Medical Chest constructed in accordance with MIL-C-16775.

3.6.8 Latches. Latches shall securely attach the parts box beneath the framed delousing unit. Four latches will be used. Latches will be spring loaded and will not impair operational handling of the unit (see 4.1).

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3.6.9 Tools and accessories. Each duster shall be complete with the following tools and accessories (see 4.1 and 6.5):

	Quantity (each)
Starter rope, engine	1
Pliers, slip joint, straight nose with cutters, 6 inches (152.4 mm) long	1
Screwdriver, flat type, 3/8 inch (9.5 mm) wide	1

3.7 Welding. Welds shall show no evidence of separation when tested as specified in 4.5.4. Fillet welds shall have equal legs and no overlaps.

3.8 Finish. Primer and enamel shall be applied to the exterior surfaces of the outfit, except for the rubber hoses and where not desirable for proper operation of the unit such as moving parts, gauges, and couplings.

3.8.1 Surface preparation. All metal surfaces to be enameled shall be thoroughly cleaned and conditioned with metal conditioner conforming to MIL-C-10578.

3.9 Marking for identification. The item covered by this specification shall be marked for identification in accordance with end item marking requirements of MIL-STD-130.

3.10 Operator's manual. Contractor shall prepare a type 14-P Operator's, Organizational, Direct and General Support, Maintenance Manual with Repair Parts List: Delousing Outfit, Power Driven; Gasoline Engine, 4-cycle, Air Cooled, 3600 RPM, 1.5/2.5 BHP; 6 Dusting Guns, Detachable Manifold in accordance with MIL-M-38784. The reading level for the manual shall be 7th grade ± 1 year. The contractor shall provide a verified manual for each delousing outfit produced (see 4.1 and 6.3).

3.11 Workmanship. The finished duster shall be clean and all sharp burrs, slivers, or splinters shall be removed. All surfaces to be soldered shall be thoroughly cleaned prior to soldering. Soldering joints shall be smooth, sound, and clean. All bolts, nuts, screws, and studs shall be drawn tight. Finish shall be smooth and free from blisters, peeling, chipping, or areas of no film. There will be no indication of rust, corrosion, scale, or dirt under, or in, finish. Welding shall be free from slag, inclusion, undercuts, incomplete sections, cracks, fractures, or incomplete fusion. Scale and flux deposits shall be removed (see 4.1).

4. QUALITY ASSURANCE PROVISIONS

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4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The supplier is responsible for ensuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards, and drawings, as applicable.

4.1.3 Disassembly inspection. Failure of any test or examination shall be cause for disassembly, in the presence of a Government representative, of the preproduction model to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected and shall be cause for rejection of the first article. Reassembly with replacement parts and retesting shall be the responsibility of the supplier.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection. The first article shall be examined for defects in table I, dimensions and weight specified, and shall be tested in accordance with 4.5.1, 4.5.2, 4.5.3, 4.5.4, and 4.5.6. The presence of any visual defect, any dimension or weight not within the specified requirements,

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4.3 (cont'd)

or failure to pass the tests shall be cause for rejection of the first article.

TABLE I. First Article Test Defects

Examine	Defect
Finish:	Type of finish not as specified. Finish applied where not desirable for proper operation of the unit. Finish blistered, peeled, chipped, or area of no film. Not finished where required. Finish not continuous, smooth and adherent (i.e, crack, runs, drips, lap, creep, orange peel, streaks, wrinkles, or flaky). Indication of rust, corrosion, or scale under finish. Separation of color, or finish rough (i.e., lint, dirt, grease, or other foreign matter embedded in finish).
Design:	Design characteristic not in accordance with the specified requirements.
Construction and workmanship (generally applicable components and assemblies):	Part missing or not type specified. Fractured, split, punctured, dented bowed, or malformed. Part misplaced, loose, or not in proper alignment. Sharp burr, sliver or projection. Operation omitted or not properly performed. Functioning component that requires abnormal force to operate. Not connected or joined as specified. Adjustable assembly that cannot be adjustef. Adjustable assembly not properly adjusted to perform the function intended. Component not readily accessible for servicing where specified.
Welding:	Not type specified, missing, incomplete, burn holes, cracked, fractured, or not fused. Slag inclusion, undercut, not smooth or uniform. Scale or flux deposits not removed. Poor surface appearance of weld as from craters, piling up of deposited metal, evidence that weld has not been ground or filled where required, or grinding or filling not neatly accomplished.

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TABLE I. First Article Test Defects

Examine	Defect
Bolts, nuts, screws, studs and other types of fasteners:	Missing, broken, stripped, fractured. Loose.
Soldering (when applicable):	Missing, not adherent or is incomplete. Not clean (flux or flux residue not removed), not smooth (surfaces not finished neatly) or pin holes in solder.
Lubrication (when applicable):	Not lubricated where required.
Gages, valves, couplings:	Not properly mounted.
Marking identification:	Missing, incomplete, not legible, not specified type or size, not in proper location, not affixed in the prescribed manner, plate not fabricated of the specified material, not stenciled, stamped, embossed, printed, lithographed, etc., as applicable.
Instruction plate:	Missing, incomplete, not legible, not in proper location.

4.4 Quality conformance inspection. Except otherwise specified herein, sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1.2, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.4.1.1 Testing of hoses. Testing of the hoses shall be performed on specimens of hose. The lot shall be all hose manufactured under the same process offered for inspection at one time. The sample unit shall be one specimen cut from the hose lot. One sample unit shall be subjected to each

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4.4.1.1 (cont'd)

test. The sample units shall be tested for burst strength and ultimate elongation in accordance with Method 10011 and Method 4121, respectively, of FED-STD-601. Any nonconformance to the requirements of 3.6.4 shall constitute failure of the tests and rejection of the lot.

4.4.1.2 Testing of spray gun discharge capacity. Each spray gun shall be tested as specified in 4.5.6. Any nonconformance to the requirements of 3.4.4 shall be cause for rejection of the spray gun.

4.4.2 In-process inspection. Examination shall be made of the following operations to establish conformance to specified requirements. Whenever nonconformance is noted, correction shall be made to the items affected and to the operation.

4.4.2.1 Visual examination of operations.

a. Metal surfaces cleaned and conditioned prior to painting as specified (see 3.8.1).

b. Metal surfaces primed as specified (see 3.4.6).

4.4.2.2 In-process testing.

4.4.2.2.1 Manifold test. Each manifold, after fabrication, and prior to application of finish shall be tested in accordance with 4.5.1. Any defect shall be cause for rejection.

4.4.2.2.2 Compressor test. One compressor from each lot manufactured under the same process shall be tested in accordance with 4.5.4. Failure to pass the test shall be cause for rejection of the lot of compressors manufactured under the same process.

4.4.3 End item inspection. A lot shall consist of all dusters offered for inspection at one time. The sample unit for these inspections shall be a completely fabricated duster (see 3.1).

4.4.3.1 Visual examination. The completely assembled end item shall be examined for defects classified in table I. The inspection level shall be II, and the presence of any defect shall be cause for rejection (see 3.1).

4.4.3.2 Dimensional and weight examination. Inspection shall be made of the end item for compliance with dimensions, and weight specified. Any noncompliance with a specified requirement shall constitute a defect. The inspection level shall be S-2, and the presence of any defect shall be cause for rejection (see 3.1).

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4.4.3.3 Testing of the end item. Each production unit shall be tested as specified in 4.5.2 and 4.5.3. Failure to pass these tests shall be cause for rejection of the unit.

4.4.4 Packaging inspection. An examination shall be made to determine that preservation-packaging, packing, and marking comply with requirements of section 5. Defects shall be as listed in table II. The sample unit shall be one shipping container fully packaged, except it shall not be sealed. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2, and the presence of any defect shall be cause for rejection.

TABLE II. Packaging Defects

Examine	Defect
Markings:	Missing; incorrect; illegible; of improper size, location, sequence, or method of application.
Preservation:	Preservative improperly applied or missing.
Materials:	Any nonconforming component; component missing or damaged.
Workmanship:	Inadequate application of components such as components of parts box not adequately cushioned and immobilized. Bulging or distorted container.

4.4.4.1 Examination of shipping containers. When shipping containers are required to be in accordance with MIL-C-52950, examination for defects in closure and strapping shall be in accordance with the appendix of that specification.

4.5 Tests.

4.5.1 Manifold. After complete fabrication of the manifold, but prior to application of finish or preservative and after plugging of all openings, the manifold shall be filled with compressed air at a pressure of 150 psig (1035 kPa) and fully submerged in water to determine compliance with 3.6.3.

4.5.2 Connections. The compressor shall be operated and permitted to build up an air pressure of 30 psig (207 kPa). All connections, including air lines, shall be tested for air leaks by the application of water containing a 20 percent solution of soap to determine compliance with the requirements

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850 F and 1350 F while testing. The slurry shall have the following pro-

4.5.2 (cont'd)

specified in 3.6, 3.6.3, and 3.6.4. The test may be performed during the operating test specified in 4.5.3.

4.5.3 Operating test. The complete unit, without a load of insecticide, shall be given an operational test of 2 hours duration. Any nonconformance of mechanical equipment to function as specified in 3.5 shall constitute failure of this test.

4.5.4 Endurance test. The compressor, with its gasoline engine, shall be subjected to 200 hours of continual operation at full-rated capacity to determine compliance with the requirements specified in 3.6 and 3.7. The 200 hours of operation may be in increments of not less than 20 hours per day, except for interruptions required for normal servicing of the equipment. Should a major breakdown occur, the endurance test shall be rerun. A major breakdown is defined as a breakdown requiring more than 1 hour during operation for any repair including all contingencies. During the test, the following data shall be recorded at 1-hour intervals; and any nonconformance to the requirements of 3.6 and 3.7 shall constitute failure of the test:

a. Back pressure of gasoline engine exhaust system [measured 2 inches (50.8 mm) from the exhaust manifold (flange)].

b. Engine speed.

c. Compressor crank shaft speed.

d. Engine-oil sump temperature.

e. Engine cylinder head temperature.

f. Compressor discharge pressure.

g. Compressor discharge temperature.

h. Compressor capacity (corrected).

i. Hydrocarbon content of discharge air.

Failure of the compressor to operate continually at rated capacity for 200 hours without a major repair; failure of the compressor to discharge oil-free air; failure of the safety valve to operate as specified in 3.6.3.2; or back pressure on the exhaust system in excess of 8 inches (203 mm) of water shall constitute failure of this test.

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4.5.5 Capacity test. The compressor component shall be driven by a direct-coupled dynamometer or a calibrated electric motor. The compressor activity shall be determined by the ASME Low Pressure Nozzle Method of the Power Test Code for Displacement Compressors, Vacuum Pumps, and Blowers utilizing the following formula:

$$Ga = \frac{2.557CDn2\ Tl\ (Pb\ p)}{Pa\qquad (T2)}\ 1/2.$$

Ga = Volumetric rate of flow cubic feet per minute (c.f.m.).

Tl = Absolute temperature at compressor intake Fahrenheit (F).

NOTE: Ambient temperature shall be determined at engine crankshaft height 3 to 5 feet from the compressor.

Pa = Absolute pressure at compressor intake, pounds per square inch (psi).
C = Nozzle coefficient of discharge.
Dn = Nozzle throat diameter (inches).
Pb = Barometric pressure (inches Hg).
T^2 = Absolute temperature, upstream of the nozzle (^oF).
p = Differential pressure across the nozzle (inches water).

The compressor speed required to produce rated capacity shall be recorded. A capacity of less than 4.2 c.f.m. (0.0030 m^3/s) at 30 psig (207 kPa), or horsepower required in excess of 1.5 shall constitute failure of the test. Other test methods by the industry are acceptable, providing results conform to the above specified test.

4.5.6 Spray gun discharge capacity test. The spray gun shall be weighed and connected to an inlet air pressure of 30 psig after filling the container to the fill line and attaching the container to the gun. With the gun in the horizontal position, the amount of talcum powder discharged by depressing the trigger shall be weighed. This test procedure shall be repeated 10 times. Nonconformance to the discharge requirements of 4.3 shall constitute failure of this test.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial as specified.

5.1.1 Level A. Preservation-packing shall be in accordance with the method and material requirements of MIL-P-116 as hereinafter specified. Where tape is specified, it shall conform to type III or IV, class 1 of PPP-T-60.

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5.1.1.1 Cleaning and drying. All exposed, uncoated ferrous metal surfaces of each duster shall be cleaned according to process C-1 and thoroughly dried.

5.1.1.2 Preservation Application.

5.1.1.2.1 Surfaces. Surfaces, cleaned and dried as specified and that do not contact other surfaces in operation, shall be coated with type P-1 preservative. Other cleaned surfaces shall be coated with type P-6 preservative. Surfaces coated with type P-6 preservative which might be removed by contact with packing materials shall be covered or wrapped with greaseproof barrier material conforming to type II, grade A of MIL-B-121, and the wrap secured with tape.

5.1.1.2.2 Gasoline engine and air compressor. The gasoline engine shall be preserved in accordance with level A requirements for type II, Method I of MIL-E-10062; except the air compressor shall not be preserved. All openings on the compressor shall be sealed with tape.

5.1.1.2.3 Components. Both ends of each spray hose shall be sealed and thoroughly covered with tape. The six discharge tubes shall be disassembled from the spray guns, wrapped together, and secured with tape or twine. The two openings into each spray gun shall be sealed with tape. Each respirator shall be preserved by method C-1 or placed in an individual polyethylene bag. The polyethylene film shall be minimum 0.0002-inch (.05 mm) thick. The mouth end of the bag, when prepared for closure, shall extend a minimum of 3 inches (76.2 mm) above the packed item. Closure of the bag shall be effected by heat sealing.

5.1.1.3 Unit Packing. The six spray hoses, six spray guns, six plastic containers, one manifold, equipment manual, discharge tubes for spray guns, and respirators, tools and accessories, preserved as specified in 5.1.1.2, shall be packed in the appropriate compartments of the parts box conforming to Drawing 60108FD, 5-13-1919. The spray guns, plastic containers, and manifold shall be cushioned with 2-pound polyurethane foam, charcoal color.

5.1.2 Commercial. Dusters shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Level A packing. Each duster, preserved as specified in 5.1, with the parts box attached in its position below the outfit, shall be packed in a snug-fitting crate conforming to type I, style A of MIL-C-52950. The duster shall be anchored and protected with a full-height shroud in accordance with the appendix of MIL-C-52950.

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5.2.2 Level B packing. Each duster, preserved as specified in 5.1, shall be packed as specified in 5.2.1; except a shroud shall not be required.

5.2.3 Commercial packing. Each duster, preserved as specified in 5.1, shall be packed in accordance with ASTM D3951.

5.3 Marking. In addition to any special marking required by the contract, packs and shipping containers shall be marked in accordance with MIL-STD-129 or ASTM D3951, as applicable.

6. NOTES

6.1 Intended use. The duster covered by this specification is intended for use in the field for delousing personnel and clothing without removing the clothing from the body.

6.2. Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Time frame when a First Article is required (see 3.2).

c. Source and delivery of Government furnished equipment (see 3.4.8).

d. Selection of the applicable levels of packaging preservation and packing (see 5.1 and 5.2).

e. Type of delouser required (see 3.4.1 and 3.4.2).

6.3 Consideration of Data Requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL) incorporated into the contract. When the provisions of DoD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraph.

Paragraph No.	Data Requirement Title	Applicable DID No.
3.10	Commercial Manuals	DI-TMSS-80527 and DI-TMSS-80528

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6.3 (cont'd)

(Data Item Descriptions related to this specification and identified in section 6 will be approved and listed as such in DoD 5010.12L. Copies of Data Item Descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.4 Inspection for First Article. When a first article inspection is required, the delouser should be a first article sample. The first article shall consist of three delousers. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products who wish to rely on such production or test must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Tools and accessories. The tools and accessories may be further identified by the Federal Stock Number (NSN) listed below:

MIL-STD-Engine 1.5/2.5 BHP Model 1A08	-	NSN 2508-00-068-7510
Pliers	-	NSN 5120-00-223-7396
Screwdriver	-	NSN 5120-00-278-1280
Air brake connection	-	NSN 4730-00-595-0083
No. 2 medical chest	-	NSN 6545-00-914-3470

6.6 Disposability. One or more of the following methods shall be used to accomplish disposal of the packaging material: reuse, recycling, sanitary landfill, incineration, or pyrolysis.

6.7 Recycled material. Unless otherwise specified herein, all equipment, materials, and articles incorporated in the product covered by this specification shall be new and fabricated using materials produced from recycled materials to the maximum extent practicable without jeopardizing the intended use. The term "recycled materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically authorized (see 3.3).

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6.8 Subject term (key word) listing.

Air compressor
Delouser
Delousing Nozzle Assembly
Drawing Package
Duster
Gasoline Engine Driven
Insecticide
Manifold
Mass Delousing

6.9 Changes from previous issue. Asterisks or vertical lines are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - MD
Navy - YD
Air Force - 99

Review activities:

Army - CE
Navy - MS, CG
Air Force - 84
DLA - CS
AFPMB - PM

Preparing activity:

Army - MD

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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
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	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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